Similarity & Proportions Maze

Name _____ Period ____

Start in the top-left room. Solve, go through the door with your answer, and show your work on the answer grid. Shadows are measured at the same time.

START

Solve.

$\frac{x}{4} = \frac{9}{12}$

$x = 3$

Solve.

$\frac{5}{x} = \frac{15}{6}$

$x = \frac{25}{2}$

In $\triangle ABC$, $\overline{DE} \parallel \overline{BC}$. Find EC.

$AD = 2$, $DB = 7$

$AE = 4$

$EC = \frac{8}{7}$

$\triangle JKL \sim \triangle MNO$. Find NO.

$JK = 12$, $MN = 8$

$KL = 15$

$x = \frac{16}{3}$

$x = 27$

$x = 2$

$x = 18$

$x = \frac{28}{3}$

$EC = 15$

$EC = 8$

Find the tower's height.

5-ft pole: 3-ft shadow

tower: 27-ft shadow

$k = 3$

A dilation maps $\triangle ABC$ onto $\triangle DEF$. Find the scale factor k.

$AB = 6$, $DE = 9$

$k = \frac{2}{3}$

Solve.

$\frac{x + 2}{5} = \frac{6}{3}$

$x = 8$

In $\triangle ABC$, $\overline{DE} \parallel \overline{BC}$. Find EC.

$AD = 4$, $DB = 6$

$AE = 6$

$\frac{81}{5}$ ft

29 ft

$k = \frac{3}{2}$

$EF = 15$

$x = 10$

$EC = 4$

$EC = 9$

Solve.

$\frac{7}{3} = \frac{x}{9}$

$x = \frac{27}{7}$

$\triangle ABC \sim \triangle DEF$. Find EF.

$AB = 4$, $DE = 10$

$BC = 6$

$EF = 12$

$\triangle PQR \sim \triangle XYZ$. Find QR.

$PQ = 10$, $XY = 4$

$YZ = 8$

33 ft

Find the tree's height.

6-ft man: 4-ft shadow

tree: 22-ft shadow

$x = 63$

$EF = 24$

$EF = \frac{12}{5}$

$QR = 20$

$QR = \frac{16}{5}$

24 ft

$\frac{44}{3}$ ft

A dilation maps $\triangle RST$ onto $\triangle UVW$. Find the scale factor k.

$RS = 20$, $UV = 8$

$x = 1$

Solve.

$\frac{x}{x + 3} = \frac{4}{7}$

$x = 4$

$\triangle ABC \sim \triangle DEF$. Find x.

$AB = 8$, $BC = 10$

$DE = 12$, $EF = 2x + 3$

$x = 6$

FINISH

You made it!

Similarity & Proportions Maze: Answer Grid

Name _____ Period ____

Write each answer in the order you found it, and show your work for that room.

1	Answer: _____	Work:
2	Answer: _____	Work:
3	Answer: _____	Work:
4	Answer: _____	Work:
5	Answer: _____	Work:
6	Answer: _____	Work:
7	Answer: _____	Work:
8	Answer: _____	Work:
9	Answer: _____	Work:
10	Answer: _____	Work: